

Introduction to Control Systems (ME2801)

Homework #2

The textbook exercises, designated by with the word “Nise”, are from “Problems” section at the end of the chapter in “Control System Engineering” by Norman Nise, 7th edition.

Nise Reading:

- Chapter 2, Sections 5 ad 6.
- Appendix B, ch2p9-ch2p12

Inverse Laplace Transform of Functions

1) Given the following Laplace transforms ($F(s)$), find corresponding time functions ($f(t)$). Use the MATLAB `residue` function for the partial fraction expansion.

You do not need to turn in your MATLAB scripts.

$$\begin{aligned} \text{a)} \quad F(s) &= \frac{s+5}{(s+2)(s^2+36)} \\ \text{b)} \quad G(s) &= \frac{2}{s(s^2+s+16.25)} \\ \text{c)} \quad H(s) &= \frac{1}{(s^2+9)(s^2+4s+1)} \end{aligned}$$

Transfer Functions

2) Nise 2.8

3) Nise 2.9

Transfer Function Representation in MATLAB

4) Consider each of the following transfer functions:

$$\text{a)} \quad G(s) = \frac{3}{s^3+9s^2+20s}$$

$$\text{b)} \quad H(s) = \frac{10(s+2)}{s^2(s+4)(s+5)}$$

For each expression, illustrate (submit a snippet of MATLAB script) how you would generate the transfer function object in MATLAB using the two methods:

- using the ratio of factors (the `zpk()` function)
- using the ratio of polynomials (the `tf()` function)

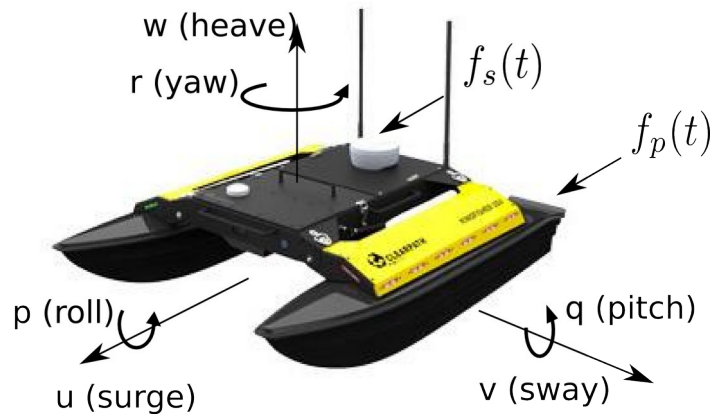
5) Nise 2.13

Note: The roots function can be used to find the roots of a polynomial. For example, to find the roots of $s^2+5s+30$ we could do the following...

```
>> roots([1 5 30])
ans =
    -2.5000 + 4.8734i
    -2.5000 - 4.8734i
```

USV Model

Consider the Kingfisher unmanned surface vessel shown in the image below.



The system has six degrees of freedom: three translations (surge, sway and heave) and three rotations (roll, pitch and yaw). The system has two inputs which are thrust forces produced by the port and starboard thrusters, $f_p(t)$ and $f_s(t)$ respectively.

- a) Consider the surge direction only. Surge velocity is $u(t)$. Find the equations of motion in the surge direction with the following assumptions:
 - The total inertia (rigid body and added mass) of the system in the surge direction is $M=36.0$ [kg]
 - The total drag of the system is a linearly related to surge velocity, with a coefficient of $D=[18.75 \text{ N/(m/s)}]$
 - The total forward thrust is $T(t)=f_s(t)+f_p(t)$ [N]
- b) Consider the scenario where the vessel starts at rest with no input. At time zero the input thrust is set to a constant 37.5 N.
 - Find an expression of the vessel speed as a function of time¹.
 - Create a graph of the vessel speed (include labels and units on the axes)
 - What is the steady-state speed of the vessel?
 - How long does it take for the vessel to reach 95% of the steady-state speed? (An estimate from the graph is acceptable.)
- c) Find the transfer function for the surge direction where the input is $T(t)$ and the output is $u(t)$.
- d) Consider the yaw direction only. The angular displacement is $r(t)$. Find the equations of motion in the yaw direction with the following assumptions:
 - The total inertia of the system in the yaw direction is $I = 8.35$ [kg m²]
 - The drag of the system is linearly related to yaw velocity, with a coefficient of $d=24$ [(Nm)/(rad/s)]
 - The total positive torque is $\tau(t)=f_s(t)-f_p(t)$ [Nm]
- d) Find the transfer function for the yaw direction where the input is $\tau(t)$ and the output is the angular displacement $r(t)$.

1 You should be able to accomplish this using what we've learned about solving LTI ODEs using Laplace transforms. It is also possible to solve this using symbolic tools (e.g., the MATLAB `dsolve()` function) or numerical tools (e.g., the MATLAB `ode45()` function), but that is not the focus of this exercise.

- e) Find the transfer function for the yaw direction where the input is $\tau(t)$ and the output is the angular velocity $\dot{r}(t)$.